

FRONTEND PRODUCT PLAN · INTERNAL DOCUMENT

Tutify

Student Module

A repo-aligned, execution-ready frontend plan for extending the current Tutify teacher module into a full student learning experience.

📁 Repo-Aligned

🎓 Student-First UX

👤 Teacher Touchpoints Mapped

🚫 No Backend Scope

⚡ Frontend Execution Ready

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MVP

PHASE 1 FOCUS



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Acceptance Checklist



Change Summary



Frontend Implementation Readiness

Context & Baseline

Why this document exists, what the repo currently supports, and the real implementation status that shapes every decision in this plan.



Document Purpose

Why this doc exists and what it is meant to do



Primary Goal

This document is a frontend product plan for building the student module of the tutify-frontend application. It extends the existing, mature teacher tooling into student-facing experiences. It is not an architecture document, not a backend spec, and not a generic product pitch. It is a concrete, role-aware, execution-ready feature plan for frontend product, design, and engineering teams.



What This Extends

Teacher module → Student module

- ▶ Teacher quiz/assignment/worksheet tools → student receive and submit surfaces
- ▶ Teacher exam builder → student exam-taking interface
- ▶ Teacher analytics → student progress view
- ▶ Teacher feedback flows → student feedback receipt and revision



What This Creates New

Student-native experiences

- ▶ Student onboarding and goal setup
- ▶ Personal homework manager (self-assigned)
- ▶ Timetable view and next-class alerts
- ▶ Grade impact calculator
- ▶ Exam day toolkit and smart notes capture
- ▶ Subject study room workspace



Current Product Baseline — Repo-Aligned

What actually exists in tutify-frontend today



Repo Reality Check

The student role currently resolves to a **/student — Coming Soon** placeholder. The student module is a greenfield build. The teacher module is the most mature area and forms the feature baseline this plan extends from.

Area	Current Status	Notes for Student Plan
Teacher — Quiz Builder	✓ Live	Student quiz-taking surface is the direct extension
Teacher — Assignment Creator	✓ Live	Student assignment hub consumes this data
Teacher — Worksheet Tool	✓ Live	Worksheet viewer/responder needed for student side
Teacher — Exam Builder	✓ Live	Student exam engine is the consumer
Teacher — Analytics Dashboard	✓ Live	Student progress view is a scoped read-only subset
Teacher — Submissions Review	~ Phase 2	Student submission status view tied to this maturity
Teacher — Deeper Analytics / Results	~ Phase 2	Student detailed performance view follows teacher-side readiness
Teacher — Candidate Management	~ Phase 2 / Preview	Student exam result view depends on this
Student Module (/student)	Placeholder	Entire student module is planned build — this doc



Planning Principle from Baseline

Phase 1 student features will only expose what teacher-side flows already support. Where teacher features are Phase 2 (e.g. submissions review, deeper analytics), the student-facing surface for those features is also deferred to Phase 2. This keeps the frontend rollout coherent and avoids building UI against APIs that don't exist yet.

SECTION 02

Student Module Vision

Experience goals, design principles, and the student value proposition — all anchored to what the repo actually supports today.

Student Experience Goals

What the student module should feel like when complete

The Student Module in One Sentence

Give students a clean, role-aware workspace where everything the teacher has created — quizzes, assignments, worksheets, exams — is surfaced clearly, actionably, and in context of the student's own learning progress and daily routine.

Role-Aware, Not Recycled

Student UI is designed for the student role — not a reskinned teacher view. Students see what's relevant to them, not administrative controls.

Teacher-Connected Always

Every piece of content a student sees traces back to a teacher action — published, assigned, or shared. The student module makes that chain explicit and transparent.

Progress Over Perfection

Phase 1 prioritizes clear task surfaces and submission flows. Deeper analytics and gamification follow teacher-side maturity — not the other way around.

Routine-Friendly

The dashboard and timetable serve the student's daily academic routine — what's due, what's next, what needs attention — at a glance.

Feedback-Ready

Submission, feedback receipt, and revision pathways are first-class — not afterthoughts. The student always knows where their work stands.

Appropriately Scoped

Students see only what they're enrolled in, assigned to, or permitted to view. No cross-class data leakage. Section/class visibility is enforced in every list view.

Design Principles

Rules the frontend team must apply when building student surfaces

#	Principle	Practical Meaning

#	Principle	Practical Meaning
1	Show status clearly	Every assignment, quiz, and exam must display its current state: not started / in progress / submitted / reviewed / graded
2	Surface the deadline	Due dates and countdowns appear on every task card — not buried in detail views
3	Submission is a first-class action	The submit button/flow is prominent, confirmed, and provides clear post-submission state
4	Feedback is always findable	Teacher comments, rubric scores, and revision requests are surfaced with a notification and permanently accessible
5	Filter by class/subject	All list views (assignments, quizzes, resources) support filtering by class/subject — students are enrolled in multiple
6	Phase-gate honestly	Features blocked by teacher-side Phase 2 readiness are shown as "coming soon" — not hidden or falsely promised

SECTION 03

Feature Map — 14 Student Modules

Each module includes: student value, teacher dependencies and touchpoints, current alignment status, MVP vs later scope, and key UI surfaces.

01



Student Onboarding

Next — New Build

Student Value

First-run setup that personalises the experience, sets learning goals, and prevents early drop-off by demonstrating value immediately.

Teacher Deps

Class/section enrollment data (teacher creates classes). Subject list from teacher setup. First assignment/quiz availability triggers "first win" moment.

Alignment

New — no onboarding exists for any role currently

MVP Scope

Profile setup → enrolled class display → goal input → 3-step tour → first task highlight

Later

Learning style quiz → AI-driven suggestions → personalised dashboard widget layout

Key UI Pages

`/student/onboarding` — multi-step wizard with progress bar, skip option, and completion state persisted to user profile

02



Student Dashboard

Next — New Build

Student Value

Daily command center: what's due today, upcoming quizzes, recently graded work, and class announcements — all in one glance.

Teacher Deps

Assignment due dates (teacher-set), quiz publish dates (teacher-set), exam schedule (teacher-set), teacher announcements.

Alignment

New surface — teacher dashboard exists; student dashboard is a role-scoped consumer view

MVP Scope

Due Soon widget · Upcoming Quizzes widget · Recently Graded widget · Class Announcements feed · Quick links to assignment hub and quiz center

Later

Streak tracker · Study goal ring · AI recommendations panel · Productivity score widget

Key UI Pages

`/student/dashboard` — responsive grid layout; each widget is individually loadable (skeleton states essential)

Student Value	Access teacher-published lessons, worksheets, and reference resources in one organised place — browsable by subject and class.
Teacher Deps	Teacher publishes/shares: lesson content, worksheets (from existing worksheet tool), PDFs, resource links. Student view is read-only consumer. Visibility is scoped to enrolled class/section.
Alignment	Existing teacher worksheet tool already creates content · New student reader/viewer surface needed
MVP Scope	Subject-filtered content list · Worksheet viewer (read-only) · PDF/resource viewer · Bookmark content · Filter by class
Later	Video lesson player · Progress tracking per lesson · AI summary generation · Offline download
Key UI Pages	/student/content · /student/content/:subjectId · /student/content/worksheet/:id

04



Assignment Hub

Next — Critical Build

Student Value

Central place to see all assigned work, track status, submit, receive feedback, and handle revisions. The highest-frequency student interaction after dashboard.

Teacher Deps

Publish: Teacher creates & assigns → student receives with due date and instructions.

Submission review: Teacher reviews (Phase 2 on teacher side) → student sees "pending review" state until then.

Feedback: Teacher posts comment/rubric → triggers student notification → student can view and respond.

Revision: Teacher requests resubmission → student re-enters submission flow.

Alignment

Existing Teacher assignment creator live · **New** Student submission UI · **Phase 2** Feedback receipt UI (depends on teacher review maturity)

MVP Scope

Assignment list (filter: all / pending / submitted / graded) · Assignment detail view with instructions + attachments · File upload submission · Status tracking badge · Due date countdown

Later

Inline rubric score display · Revision request flow · Teacher comment thread · Rich text submission editor · AI plagiarism check indicator

Status States

Not Started **In Progress (Draft)** **Submitted** **Pending Review**
Graded **Revision Requested**

Key UI Pages

/student/assignments · /student/assignments/:id ·
/student/assignments/:id/submit ·
/student/assignments/:id/feedback

05



Quiz & Assessment Center

Next — Directly Extends Teacher Tools

Student Value

Take teacher-built quizzes and exams in a focused interface. See results immediately (where teacher permits) or when released. Review answers and understand weak areas.

Teacher Deps

Quiz Builder (live): Student quiz-taking UI is the direct consumer.

Exam Builder (live): Student exam engine renders teacher exam config.

Scheduling: Teacher sets availability window → student sees countdown to open/close.

Results release: Teacher controls when results are visible → student sees "results pending" until released.

Candidates/Results (Phase 2): Detailed per-student results depend on teacher-side Phase 2 readiness.

Alignment

Existing Quiz & exam tools live on teacher side · **New** Student taking/viewing surfaces · **Phase 2** Detailed results view

MVP Scope

Upcoming quizzes/exams list · Quiz-taking interface (timer, question nav,

	submit) · Immediate score display (where allowed) · Answer review (where teacher permits) · Basic result card
Later	Topic-wise score breakdown · Time-per-question analysis · Weak concept highlight · Exam readiness score · Practice mode
Key UI Pages	<code>/student/quizzes</code> · <code>/student/quiz/:id/take</code> · <code>/student/quiz/:id/results</code> · <code>/student/exams</code> · <code>/student/exam/:id/take</code>

06



Personal Homework Manager

Next — Student-Native

Student Value	Separate from teacher-assigned work — students self-log study tasks, textbook exercises, and personal practice. Prevents overload across 6–8 subjects daily.
Teacher Deps	Minimal — subject list from enrolled classes provides subject taxonomy. Personal tasks are student-created, not teacher-initiated. However, teacher-assigned work displays alongside for load awareness.
Alignment	New — no equivalent in teacher module; fully student-native feature
MVP Scope	Quick-add task (subject + title + due date) · Today's task list with check-off · Filter by subject · Total time estimate for today's load
Later	Voice-add via mic · Photo capture from blackboard (see Module 09) · Load analyzer chart · AI redistribution suggestion · Completion rate analytics
Key UI Pages	<code>/student/homework</code> — split view: Today's tasks left, Subject breakdown right. Quick-add slide-up panel.

07



Class Timetable

Next — Student-Native

Student Value	Period-by-period weekly view of all classes. "Next class in X minutes" live countdown. Join online class button per period. Never miss a class.
Teacher Deps	Class schedule data set by teacher/admin. Subject name, teacher name, room, and online link per period come from teacher-created class configuration. Substitute and cancellation alerts come from teacher-posted announcements.
Alignment	New surface — timetable data exists in class configuration but no student-facing timetable view exists
MVP Scope	Weekly grid view (Mon–Sat) · Subject + teacher + room per period · Next class countdown · Join class button for online sessions · Class cancelled indicator
Later	Export to device calendar · Home screen widget (mobile) · Pre-class reminder push notification · Smart alert for schedule changes
Key UI Pages	<code>/student/timetable</code> — sticky top bar shows "next class" even when browsing other sections

08



Subject Study Room

Later — Phase 2

Student Value	Dedicated per-subject workspace: notes, resources, Q&A board, key formulas, and mastery snapshot — all scoped to one subject.
Teacher Deps	Teacher-shared resources and materials populate the study room. Subject

Q&A board posts are visible to teacher. Teacher may pin important posts. Formula board is student-editable but teacher can push subject-specific formula sheets.

Alignment	New — no equivalent exists; requires content sharing infrastructure from teacher side
MVP Scope	Subject resource list · Personal notes tab (per subject) · Pinned formula board · Basic Q&A thread
Later	Peer collaboration in Q&A · Teacher-highlighted answers · Mastery snapshot tied to quiz scores · AI suggested resources
Key UI Pages	/student/subjects/:id/room — tabbed layout: Resources / Notes / Q&A / Formulas

09



Smart Capture & Notes

Later — Phase 2/3

Student Value	Camera → OCR → structured digital notes. Math problem photo → step-by-step AI solution. Handwriting digitiser. Reduces friction of capturing classroom content.
Teacher Deps	Captured content can be auto-filed to the relevant subject room (Module 08). Teacher-shared formula sheets can serve as reference context for math scanning. No teacher action required to initiate capture.
Alignment	New — requires OCR/ML capability; scoped to mobile-first experience
MVP Scope	Notes rich-text editor · Manual note creation per subject · Basic file attach and image embed
Later	Camera capture + OCR → auto-note · Math photo solver · Handwriting digitiser · Auto-flashcard generation from captured notes
Key UI Pages	/student/notes · /student/notes/new · /student/notes/:id · Camera capture modal (Phase 3)

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Study Time Tracker

Later — Phase 2

Student Value	Students log study time per subject. Budget vs actual view reveals subject imbalance. Simple timer, manual entry, or session-auto-tracking.
Teacher Deps	Subject list from enrolled classes provides taxonomy. No teacher action required. Time data is private to the student unless the institution enables teacher visibility.
Alignment	New — fully student-native; no teacher-side dependency for MVP
MVP Scope	Start/stop timer per subject · Manual time entry · Daily total display · Week summary bar chart
Later	Budget-setting per subject · Budget vs actual chart · "You're spending 3× more on Math than Chemistry" insights · Correlation with quiz scores
Key UI Pages	/student/study-time — timer control top, subject breakdown below, weekly chart at bottom

11



Grade Impact Calculator

Next — High Student Value

Student Value	"What score do I need on the final to get a B?" — instantly answers the most common student question. Scenario modeler for exam planning.
Teacher Deps	Teacher sets assignment/quiz/exam weightings in class configuration → calculator pulls these automatically. If weightings aren't set, student can manually input them. Grade visibility depends on teacher releasing grades.

Alignment	New — frontend calculation tool; pulls existing grade data from teacher-released results
MVP Scope	Input: current grade + upcoming assessment weightage → Output: required score to achieve target grade. Best/worst case display. Works for one subject at a time.
Later	GPA impact across subjects · Visual grade trajectory chart · Auto-pull from teacher-released grades · Multi-subject scenario comparison
Key UI Pages	<code>/student/grade-calculator</code> — accessible from assignment hub sidebar and exam toolkit

12



Exam Day Toolkit

Next — High Student Value

Student Value	Exam countdown, pre-exam checklist, night-before revision pack, and post-exam reflection. Supports the student at the highest-stress academic moments.
Teacher Deps	Exam schedule and metadata (date, time, duration, location, allowed materials) sourced from teacher-created exam configuration. Night-before revision content pulls from teacher-published subject materials. Post-exam results tied to teacher-released grades.
Alignment	Existing Teacher exam builder provides schedule data · New Student-facing exam day UI · Phase 2 AI-generated revision pack
MVP Scope	Exam list with countdown · Exam detail card (metadata from teacher config) · Pre-exam checklist (student-editable) · Post-exam mood check-in · Expected score self-log
Later	AI "night before pack" (key topics auto-generated from subject content) · Exam readiness score · Result vs self-estimate comparison · Improvement plan generation
Key UI Pages	<code>/student/exams</code> · <code>/student/exams/:id/prepare</code> · <code>/student/exams/:id/reflect</code>

13



Teacher Interaction Hub

Next — Extends Messaging

Student Value	Structured way to interact with teachers: raise a doubt, book a meeting slot, view office hours, and track feedback received — all in one hub rather than scattered across assignments.
Teacher Deps	Doubt Tickets: Student submits → teacher receives in their notification feed → teacher responds → student notified. Office Hours: Teacher publishes office hours → student sees and books slot. Feedback History: All teacher feedback across assignments and quizzes consolidated here. Sharing: Teacher can push supplementary resources directly to student via interaction hub.
Alignment	Partial Messaging exists; structured doubt ticket and office hours are new patterns
MVP Scope	Message thread per teacher (per class) · Doubt submission form (subject + chapter + description + optional image) · Doubt status tracking (open / answered / resolved) · Feedback history aggregated view
Later	Office hours calendar · Appointment booking flow · Video call initiation · Feedback quality rating (student rates teacher response)
Key UI Pages	<code>/student/teachers</code> · <code>/student/teachers/:id</code> · <code>/student/doubts</code> · <code>/student/doubts/:id</code>



Student Value	Student-facing summary of their own performance: quiz scores over time, assignment completion rate, grades released by teacher — scoped to what the student is permitted to see.
Teacher Deps	Teacher analytics dashboard is the data source. Student view is a scoped, read-only subset. Only data the teacher has released (grades, results, feedback) is visible. Detailed analytics parity follows teacher-side Phase 2 analytics maturity.
Alignment	Existing Teacher analytics live · New Student read-only view · Phase 2 Detailed breakdown depends on teacher analytics Phase 2
MVP Scope	Quiz score history (line chart) · Assignment completion rate per subject · Released grade display per assessment · Attendance summary (if tracked)
Later	Topic-wise performance breakdown · Weak area identification · Score trend prediction · Comparison to class average (where teacher enables)
Key UI Pages	<code>/student/progress</code> · <code>/student/progress/:subjectId</code>

Teacher ↔ Student Interaction Matrix

Explicit mapping of every cross-role flow — from teacher action to student experience and back.
This is the core of what the student module must handle correctly.

Full Cross-Role Flow Map

Every significant teacher→student and student→teacher interaction in the system

Flow	Teacher Action	Student Experience	Student → Teacher Return	Phase
Assignment Publish	Creates and assigns with due date, instructions, attachments	Appears in Assignment Hub with status "Not Started" + due date countdown + notification	Student submits → teacher notified	Phase 1
Assignment Submission	Receives submission in teacher submissions review (Phase 2)	Status updates to "Submitted" → confirmation message shown	Awaits teacher review/grade	Phase 1 submit · Phase 2 review
Assignment Feedback	Posts grade + rubric score + comment via teacher review panel	Notification received → feedback view shows grade + comment + rubric breakdown	Student can reply or request clarification	Phase 2
Revision Request	Marks assignment for resubmission with note	Status changes to "Revision Requested" + notification + teacher note displayed	Student resubmits via same flow	Phase 2
Quiz Publish	Creates quiz, sets availability window, assigns to class	Appears in Quiz Center with countdown to open + duration info	—	Phase 1
Quiz Taking	Quiz auto-graded (MCQ) or pending teacher grade (open)	Student completes → immediate score for auto-graded items → "results pending" for open questions	Responses submitted to teacher gradebook	Phase 1
Quiz Results Release	Teacher releases results (controls visibility)	Notification → results view shows score, correct answers (where permitted), time taken	—	Phase 1 basic · Phase 2 detailed
Exam Schedule	Creates exam with date, time, duration, allowed materials via exam builder	Appears in Exam Day Toolkit with countdown, metadata, pre-exam checklist context	—	Phase 1

Flow	Teacher Action	Student Experience	Student → Teacher Return	Phase
Exam Sitting	Exam opens at teacher-scheduled time	Student enters exam via assessment center — full-screen, timed, question navigator	Responses submitted	Phase 1
Exam Results	Teacher releases via candidates/results (Phase 2)	Notification → result card displayed → detailed breakdown in Phase 2	—	Phase 2
Worksheet Distribute	Publishes worksheet to class via worksheet tool	Appears in Content Viewer under relevant subject — readable/interactive	Completed worksheet submission (if interactive)	Phase 1 view · Phase 2 submit
Resource Share	Uploads PDF, link, or note to subject resources	Appears in Content Viewer and Subject Study Room under subject	Student bookmarks or comments	Phase 1
Announcement Post	Posts class announcement (text, optionally with link/file)	Appears in Dashboard feed + Notification bell + subject area	Student can react/acknowledge	Phase 1
Doubt Ticket	Receives doubt → responds with text ± resource link	Student submits → sees "Open" status → notified when teacher responds → views answer	Student marks resolved or asks follow-up	Phase 1
Grade Visibility	Controls which grades are visible to student (per class settings)	Student progress view only shows teacher-released data — no grade appears until teacher publishes	—	Phase 1
Class Sharing Controls	Sets which students are in which class/section	Student only sees content, peers, and assignments from their enrolled classes	—	Phase 1



Sharing Model Clarification

"Sharing" in this system is primarily academic distribution — teacher publishes content to a class, and enrolled students receive it. Peer-to-peer resource sharing (student → student) is a Phase 3 social layer feature and is not part of the core submission and review loop above.

Phased Frontend Rollout Plan

Three phases of frontend feature delivery — each phase is scoped to what teacher-side and infrastructure capabilities actually support at that point in time.

1 Phase 1 — Student MVP

Aligned to current teacher module capabilities — buildable now

Phase 1 · Student MVP · Aligned to Existing Teacher Tools

Goal: Students can log in, find their work, submit assignments, take quizzes/exams, and communicate with teachers. Core daily workflow only.

- | | |
|---|--|
| ✓ Student Onboarding Wizard (3-step) | ✓ Student Dashboard — Due Soon + Quizzes + Announcements |
| ✓ Assignment Hub — List + Detail + File Submit + Status | ✓ Quiz Taking Interface — Timer + Nav + Submit + Basic Score |
| ✓ Exam Taking Interface — extends quiz engine | ✓ Content Viewer — Lessons + Worksheets (read-only) |
| ✓ Class Timetable — Period grid + Next class countdown | ✓ Personal Homework Manager — Quick add + Checklist |
| ✓ Grade Calculator — Manual input + Scenario output | ✓ Exam Day Toolkit — Countdown + Checklist + Post-reflect |
| ✓ Teacher Interaction — Doubt ticket + Message thread | ✓ Student Progress View — Quiz scores + Completion rate |
| ✓ Notifications — Due date, publish, feedback released | ✓ Class/subject scoped views on all list pages |



Phase 1 Constraints

Submission review feedback (teacher→student comments), detailed rubric display, and exam result detail pages are shown as "results pending" or "coming soon" states in Phase 1. They are not built yet because teacher-side review/results features are Phase 2 on the teacher module.

2 Phase 2 — Submission & Review Parity

Phase 2 · Submission & Feedback Parity · Teacher Phase 2 Dependent

Goal: Full two-way submission-feedback loop. Student sees detailed teacher feedback, rubric scores, and revision requests. Deeper analytics surfaces open.

- | | |
|---|--|
| ✓ Teacher feedback receipt UI (comments + rubric) | ✓ Revision request flow — resubmission path |
| ✓ Detailed quiz/exam result breakdown | ✓ Answer review page (per teacher permission) |
| ✓ Topic-wise quiz score breakdown | ✓ Worksheet interactive submission |
| ✓ Subject Study Room — full feature | ✓ Study Time Tracker — budget vs actual charts |
| ✓ Notes system — rich text per subject | ✓ Teacher Interaction Hub — office hours + booking |
| ✓ Progress View — detailed per subject | ✓ Grade auto-pull from teacher-released results |
| ✓ Peer resource sharing (class-scoped) | ✓ Onboarding — learning style quiz + personalisation |

3 Phase 3 — Engagement & Collaboration Layer

Advanced features — not required for core student workflow

Phase 3 · Engagement & Collaboration · Advanced Layer

Goal: Make learning engaging beyond compliance. Social, collaborative, and AI-assisted features that extend the core workflow.

- | | |
|---|---------------------------------------|
| ✓ Smart Capture — Camera OCR + Math scanner | ✓ AI study suggestions from content |
| ✓ Gamification — streaks, XP, badges | ✓ Social learning feed (class-scoped) |
| ✓ Project workspace + Kanban | ✓ Real-time collaborative notes |
| ✓ Long-term academic goal planner | ✓ AI Night Before Pack (exam prep) |
| ✓ Study Music / Focus mode UI | ✓ Extracurricular manager |
| ✓ Student progress sharing with parents | ✓ Emergency academic support button |

Appendices

Out-of-scope declaration, acceptance checklist, change summary, and frontend implementation readiness guidance.

Out-of-Scope Declaration

What this document deliberately does not cover

Category	Excluded Content	Reason
Backend Architecture	API design, microservices, service mesh, REST endpoint specifications	Backend spec is a separate document
Database	Data models, schema design, migrations, ORM config	Not frontend scope
Infrastructure	Cloud config, Kubernetes, CDN, CI/CD pipelines, multi-region deploy	DevOps/infra team scope
Security Architecture	Auth flows internals, token strategy, GDPR data handling at storage layer	Backend/security team scope
AI/ML Systems	Model training, inference infra, embedding pipelines	AI team scope; frontend calls APIs only
Analytics Backend	Event pipeline, ClickHouse, Kafka, cohort computation	Data engineering scope
Financial/Payments	Fee processing, payment gateways, invoicing backend	Out of student module scope for now
Marketplace/Plugin Ecosystem	Plugin registry, vendor system, revenue sharing	Future product scope

Acceptance Checklist

Criteria this document must satisfy

- ✓ Repo-aligned — features mapped to current tutify-frontend teacher module state
- ✓ Teacher touchpoints explicit — every module includes teacher dependency and interaction behavior
- ✓ No backend sections — architecture, DB, infra content excluded throughout
- ✓ Phase-gated honestly — Phase 2 teacher features acknowledged as blockers for related student features
- ✓ Submission lifecycle defined — not started / submitted / pending review / graded / revision states mapped
- ✓ Feedback loop defined — teacher comment, rubric, and revision request surfaces described

- ✓ Sharing model explicit — academic distribution vs peer sharing clearly differentiated
- ✓ Assessment surfaces defined — quiz/exam taking interfaces tied to teacher setup
- ✓ Class/section visibility stated — all list views scoped to enrolled classes
- ✓ Implementation-ready — key UI pages named with routes, MVP vs later scopes defined
- ✓ Now / Next / Later labelling — status labels used throughout modules
- ✓ Frontend-only language — no backend terminology used in feature descriptions

Change Summary

What changed from the source document to produce this rewrite



Removed

Content excluded from this plan

- ▶ All backend/microservices architecture sections
- ▶ Database model reference tables (100+ models)
- ▶ Cloud/infra/DevOps stack detail
- ▶ Security architecture deep-dives
- ▶ AI/ML pipeline and training references
- ▶ Analytics backend pipeline diagrams
- ▶ Enterprise marketplace/plugin ecosystem detail
- ▶ Generic mega-platform stat claims
- ▶ Financial/payment system deep-dive
- ▶ International compliance architecture



Refocused

Content reframed for this context

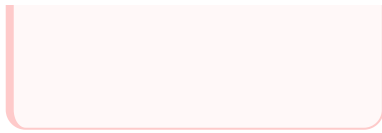
- ▶ Student features rewritten as repo extensions, not standalone ideas
- ▶ Each module anchored to teacher-side capability status
- ▶ Phase gates tied to teacher module maturity, not generic timelines
- ▶ Analytics coverage reframed as student-facing subset of teacher analytics
- ▶ Onboarding redesigned around class enrollment reality
- ▶ Grade calculator tied to teacher-released grade visibility controls
- ▶ Exam toolkit tied to teacher exam builder config
- ▶ "Social sharing" refocused as academic distribution model first



Added

Teacher-related coverage now explicit

- ▶ Teacher dependency block on every module
- ▶ Explicit submission lifecycle states across all modules
- ▶ Feedback loop UI descriptions (comment, rubric, revision)
- ▶ Teacher ↔ Student Interaction Matrix (16 flows)
- ▶ Phase 2 blocker acknowledgements per module
- ▶ Class/section visibility statement per module
- ▶ Named routes for all key UI pages
- ▶ Sharing model clarification (academic vs social)
- ▶ Current repo baseline table with status column
- ▶ Frontend implementation



readiness section
(below)

⚡ Frontend Implementation Readiness

Actionable clarity for product, design, and frontend teams — what to do next

Start Here — Phase 1 Buildable Now

All Phase 1 features in this plan are implementable against existing teacher-side APIs. No backend additions are required to begin. The student module is a greenfield build consuming existing data endpoints.



Frontend Team — Build Order

Recommended sequence for Phase 1

- ▶ **Week 1–2:** Routing setup · Auth guard for student role · Onboarding wizard · Dashboard skeleton
- ▶ **Week 3–4:** Assignment Hub — list, detail, submit flow, status states
- ▶ **Week 5–6:** Quiz taking engine · Exam taking engine (extends quiz)
- ▶ **Week 7–8:** Content Viewer (lessons + worksheets read-only) · Timetable
- ▶ **Week 9–10:** Homework Manager · Grade Calculator · Exam Toolkit
- ▶ **Week 11–12:** Teacher Interaction (doubts + messages) · Progress View · Notifications



Design Team — Priority Screens

Design these first to unblock dev

- ▶ **P0:** Student dashboard layout (widget system)
- ▶ **P0:** Assignment list + detail + submission flow
- ▶ **P0:** Quiz taking interface (full-screen, timer, nav)
- ▶ **P0:** Onboarding wizard (3 steps + completion)
- ▶ **P1:** Timetable weekly grid
- ▶ **P1:** Notification patterns (bell, toast, in-page)
- ▶ **P1:** Status badge system (assignment/quiz states)
- ▶ **P2:** Feedback receipt page · Rubric display



What to Build Against Today

Existing teacher APIs to consume

- ▶ Teacher-created assignments → student assignment list endpoint
- ▶ Teacher quiz config → quiz taking render engine
- ▶ Teacher exam config → exam taking render engine



What to Show as Pending (Phase 2)

Don't build UI against unavailable APIs

- ▶ Teacher inline feedback → show "Feedback not yet available" state
- ▶ Rubric score display → placeholder until teacher review goes live
- ▶ Detailed exam results → "Results will

- ▶ Teacher worksheet → worksheet viewer
- ▶ Teacher class enrollment → timetable + scoped views
- ▶ Teacher announcements → dashboard feed
- ▶ Teacher-released grades → progress view + grade calculator pre-fill

appear when released" state

- ▶ Worksheet submission → show read-only until submission API ready
- ▶ Candidate/results detail → scoped to Phase 2 teacher maturity
- ▶ Deep analytics breakdown → "More detail coming" with basic summary



Readiness Verdict

Product: Feature map and interaction matrix are decision-ready. Phase gates are clear. No ambiguity about what is in Phase 1 vs 2 vs 3.

Design: Screen list is defined per module. Start with P0 screens above. Status badge system and notification patterns need a design system addition.

Frontend Engineering: Routes are named. Teacher API surfaces to consume are identified. Phase 2 "pending" states prevent over-building. Begin with routing scaffold and onboarding flow.

Tutify · Student Module Frontend Plan

TUTIFY-FRONTEND · V1.0 · REPO-ALIGNED · FRONTEND SCOPE ONLY

14 Student Modules

3 Rollout Phases

Teacher Touchpoints Mapped

Frontend Execution Ready

Repo-Aligned